Week 11: Chip-firing with a sink

Adult guide built from the problem files · Unpiloted. Counter fit on the 44 mm circles is unrehearsed.

This view repeats, by problem, the answers of the print guide, week-11-facilitator.pdf, whose source is editable/facilitator-src/build_guide.py; its page references are to that guide. The overview, preparation, launch, proofs and hints are there. Change both together; check_guide_copy.py compares them.

Problems and routes

Problem	K–1	2–3	4–5	route
Two orders from each start on the triangle	P1	P1	P1	first
Every finish from 4 chips, then from 5	P2			first
Two orders from each start on the four-cycle	P3	P2		first
Six-chip starts that finish at one and one		P3		first
Every placement where no circle can share	P4			return
Batches together or one at a time			P2	first
Return and merging when adding at A			P3	first
Order on the board with an extra line		P4	P4	return
Adding chips while sharing, or in batches	P5	P5		return
Adding one chip at a time on the triangle	P6	P6		return
Why sharing must stop on the four-cycle			P5	return
Why every order gives the same piles and counts			P6	return

First route. K–1 P1–P3; Grades 2–3 P1–P3; Grades 4–5 P1–P3. **Return visits.** K–1 P4–P6; Grades 2–3 P4–P6; Grades 4–5 P4–P6.

Key, one entry per problem

Two orders from each start on the triangle

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. On the triangle with a sink, every legal order from a start gives the same final piles and the same number of shares at each circle.

K–1: Triangle starts. Changing legal move order cannot change the finish. The answer rows are:

Start	Finish	Shares at each circle	One legal order
$(2,2)$	$(1,1)$	$(1,1)$	AB
$(3,2)$	$(1,0)$	$(2,2)$	ABAB
$(2,3)$	$(0,1)$	$(2,2)$	ABBA
$(3,3)$	$(1,1)$	$(2,2)$	ABAB

The share-count column is adult support; children are only asked about final piles. These trials deliberately give three different finishes across different starts. Each start is printed on two rows, one for each order, so that children compare two runs of the same start rather than different starts.

Grades 2–3: Triangle finishes and sharing counts are both independent of legal order, for the same start.

Start	Finish	Shares at each circle	One legal order
$(2,2)$	$(1,1)$	$(1,1)$	AB
$(4,0)$	$(1,0)$	$(2,1)$	AAB
$(3,3)$	$(1,1)$	$(2,2)$	ABAB

Two examples of orders are AB and BA for $(2,2)$, and ABAB and BABA for $(3,3)$. The start $(4,0)$ has only one complete legal word, AAB; the printed "if possible" matters. A cannot skip a legal forced move by inventing a second order.

Grades 4–5: For one fixed start, final piles cannot agree while share counts differ. Here are all four records:

Start	Finish	Shares at each circle	Chips in sink
$(2,2)$	$(1,1)$	$(1,1)$	2
$(4,0)$	$(1,0)$	$(2,1)$	3
$(3,3)$	$(1,1)$	$(2,2)$	4
$(5,4)$	$(1,0)$	$(4,4)$	8

Use AB/BA; AAB only; ABAB/BABA; and AABABBAB/BBAABAAB, respectively. Across *different* starts, equal finishes with different share counts do occur: $(2,2)$ and $(3,3)$ are examples. Preserve the printed fixed-start qualification.

K–1: If a child moves just one chip, touch both lines from the firing circle and send a chip along each. For an order comparison, restore the identical start first.

Grades 2–3: Keep the table routine from page 4: one child moves chips and the other checks each line and records the firing word; then exchange roles. After stopping, count its letters to recover the sharing counts. A count at A means the number of times A shared, not chips remaining at A.

Every finish from 4 chips, then from 5

K–1 P2 · first route

Idea. Each total of 4 or 5 chips on the triangle reaches exactly the three finishes (1,0), (0,1) and (1,1); listing A = 0 to the total covers every start.

Distribute 4 chips altogether, then 5 altogether. Each total has exactly three possible finishes: $(1,0),(0,1),(1,1)$. The following is a complete check, with A increasing by one.

4 chip start	Finish	5 chip start	Finish
$(0,4)$	$(0,1)$	$(0,5)$	$(1,0)$
$(1,3)$	$(1,0)$	$(1,4)$	$(1,1)$
$(2,2)$	$(1,1)$	$(2,3)$	$(0,1)$
$(3,1)$	$(0,1)$	$(3,2)$	$(1,0)$
$(4,0)$	$(1,0)$	$(4,1)$	$(1,1)$
		$(5,0)$	$(0,1)$

Completeness. A can hold 0 through the total; B must hold the rest. This lists every start, including empty circles. All legal orders from any listed start share its finish. With four or five chips, $(0,0)$ never occurs.

Keep the same four counters and move one from B to A between trials. Let the child decide how to keep the different finishes.

Two orders from each start on the four-cycle

K–1 P3 · Grades 2–3 P2 · first route · wording differs by band

Idea. On the four-cycle, where B sends only to A and C, every legal order from a start still gives one finish and one count vector.

K–1: Four-cycle starts. All legal orders from a fixed start agree:

Start	Finish	Shares at each circle	One legal order
$(0,4,0)$	$(1,0,1)$	$(1,3,1)$	BBACB
$(2,2,0)$	$(1,1,1)$	$(1,1,0)$	AB
$(0,2,2)$	$(1,1,1)$	$(0,1,1)$	BC
$(2,0,2)$	$(1,0,1)$	$(1,1,1)$	ACB

Grades 2–3: Four-cycle starts have these exact records:

Start	Finish	Shares at each circle	One legal order
$(0,4,0)$	$(1,0,1)$	$(1,3,1)$	BBACB
$(2,1,2)$	$(1,1,1)$	$(1,1,1)$	ABC
$(2,4,2)$	$(1,0,1)$	$(3,5,3)$	ABBABCCBACB

Other complete words are BBCAB, CBA, and CBBCBAABCAB, respectively. Every row duplicated on the student page should end with the same finish and count vector.

K–1: B sends to A and C, not to the sink. On $(0,4,0)$, B may share twice before either neighbor shares. Ask where every chip went if the total appears to change.

Grades 2–3: If records disagree, replay the two words slowly. Check for an illegal early firing or a missed chip along the B–C edge before discussing a counterexample.

Six-chip starts that finish at one and one

Grades 2–3 P3 · first route

Idea. Of the seven ways to place 6 chips on the triangle, exactly (0,6), (3,3) and (6,0) finish at (1,1); choosing A from 0 to 6 forces B, so the list is complete.

Exactly three starts of total six finish at $(1,1)$: $(0,6), (3,3), (6,0)$.

A initially	0	1	2	3	4	5	6
B initially	6	5	4	3	2	1	0
Finish	$(1,1)$	$(0,1)$	$(1,0)$	$(1,1)$	$(0,1)$	$(1,0)$	$(1,1)$

Why complete. There are seven possible values of A, 0 through 6. Once A is chosen, B is forced. Checking these seven starts leaves none untested. Order independence means no unsuccessful start can be rescued by a new firing order.

If the child finds $(3,3)$ and stops, ask whether an empty circle is permitted at the start. If the list is disorganized, ask how one could be certain it includes every possible A pile.

Every placement where no circle can share

K–1 P4 · return route

Idea. A stable four-cycle placement has 0 or 1 chip at each circle: eight placements, and none uses 4 chips because three circles hold at most one each.

Every stable four-cycle state has 0 or 1 chip at each circle. The complete set of eight is

$$(0,0,0),\ (0,0,1),\ (0,1,0),\ (0,1,1),\qquad (1,0,0),\ (1,0,1),\ (1,1,0),\ (1,1,1).$$

There cannot be a stable placement of four chips: three circles hold at most one each. This is a capacity proof, not merely a failed search.

Ask how many chips one circle can hold without sharing. Once the child has examples, sort them by whether A is empty; each half has the four B/C choices. If a child stops at seven, ask whether a board with no chips counts; the eighth way is $(0,0,0)$. The page has ten rows so that it does not give the count away.

Batches together or one at a time

Grades 4–5 P2 · first route

Idea. Adding two batches together, or finishing one before adding the other, ends at the same finish: S(S(x)+y) = S(x+y).

Together, first batch then second, and second then first all give the same finish for each row group:

Batches	Together	First then second	Second then first
4 at B; 2 at A	$(1,1,1)$	$(1,1,1)$	$(1,1,1)$
3 at A; 3 at C	$(1,0,1)$	$(1,0,1)$	$(1,0,1)$

For the first pair, intermediate finishes are $(1,0,1)$ or $(0,1,0)$, as appropriate. For the second pair, $(3,0,0) \to (1,1,0)$, then $(1,1,3) \to (1,0,1)$; reversing A/C gives $(0,1,1)$ then $(3,1,1) \to (1,0,1)$. This does not claim the intermediate states agree.

Compare only after the same total additions have arrived and sharing is finished.

Return and merging when adding at A

Grades 4–5 P3 · first route

Idea. Adding one chip at A and finishing cycles (1,0), (0,1), (1,1) with period three; the empty start never returns, and (0,0) and (1,1) merge after one addition.

The four stable starts have the following A-addition updates:

Start	After 1	After 2	After 3	After 4
$(0,0)$	$(1,0)$	$(0,1)$	$(1,1)$	$(1,0)$
$(1,0)$	$(0,1)$	$(1,1)$	$(1,0)$	$(0,1)$
$(0,1)$	$(1,1)$	$(1,0)$	$(0,1)$	$(1,1)$
$(1,1)$	$(1,0)$	$(0,1)$	$(1,1)$	$(1,0)$

The three nonempty starts return after three additions and then at multiples of three. The empty start never returns. Starts $(0,0)$ and $(1,1)$ merge after one addition at $(1,0)$: the operation cannot be reversed uniquely. The state diagram on page 7 is complete.

Ask which arrows could be reversed without making a choice. Do not label all stable states a group.

Order on the board with an extra line

Grades 2–3 P4 · Grades 4–5 P4 · return route · wording differs by band

Idea. With the A-C line, A and C need 3 chips and B still needs 2; some finishes change, but every legal order from a start still agrees.

Grades 2–3: On the extra-line board, A and C have degree 3, while B has degree 2. The complete answer rows are:

Start	Finish	Shares at each circle	One legal order
$(3,0,3)$	$(2,0,2)$	$(1,1,1)$	ACB
$(0,6,0)$	$(2,0,2)$	$(1,4,1)$	BBBACB
$(2,2,2)$	$(2,0,2)$	$(1,2,1)$	BACB
$(4,1,2)$	$(2,1,0)$	$(2,2,2)$	ABCABC

Alternative legal orders, in the same row order, are CAB, BBBCAB, BCAB, and ACBABC. The added line changes thresholds and some finishes, but not the order-independence conclusion.

Grades 4–5: Extra-line answers: $(3,0,3) \to (2,0,2)$ with counts $(1,1,1)$; $(0,6,0) \to (2,0,2)$ with counts $(1,4,1)$; $(4,1,2) \to (2,1,0)$ with counts $(2,2,2)$. Orders from page 9 give two trials for each. Legal order still changes neither record.

Grades 2–3: Before the first trial, let the child point to every destination for A's three outgoing chips. If they fire A at two chips, ask for a chip to match its third line.

Adding chips while sharing, or in batches

K–1 P5 · Grades 2–3 P5 · return route · wording differs by band

Idea. Once all the prescribed chips have arrived and sharing is finished, the finish does not depend on when the chips were added.

K–1: For every timing of additions and legal sharing, $(3,3)$ finishes at $(1,1)$, and $(2,4)$ finishes at $(1,0)$. All three repeated rows for each total have the same answer. In a batch of 3 at A then 3 at B: $(3,0) \to (1,1)$, then $(1,4) \to (1,1)$. Reversing the batch order agrees. For 2 at A then 4 at B: $(2,0) \to (0,1)$, then $(0,5) \to (1,0)$. Reversing gives $(0,4) \to (0,1)$, then $(2,1) \to (1,0)$.

Grades 2–3: The batches are 2 at A and 4 at B on the four cycle. All three timing choices finish at $(1,1,1)$. Together, $(2,4,0)$ has total firing counts $(2,3,1)$. For A's batch first, $(2,0,0) \to (0,1,0)$, then $(0,5,0) \to (1,1,1)$. For B's batch first, $(0,4,0) \to (1,0,1)$, then $(3,0,1) \to (1,1,1)$.

K–1: Reserve the not-yet-added chips off the board, away from the sink. Stop only after all scheduled chips have been added and no circle can share.

Grades 2–3: Have the child name what is held back at each stage. Empty the sink between independent trials, never between the two batches of one trial.

Adding one chip at a time on the triangle

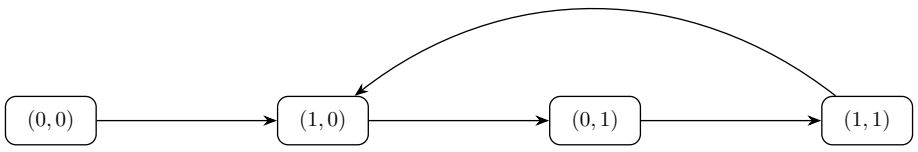
K–1 P6 · Grades 2–3 P6 · return route · wording differs by band

Idea. Adding one chip at A and finishing cycles through (1,0), (0,1), (1,1); the empty board never returns; adding at B runs the same cycle the other way.

K–1: After adding 1 through 11 chips at A, the finishes are

$$(1,0), (0,1), (1,1), (1,0), (0,1), (1,1), (1,0), (0,1), (1,1), (1,0), (0,1).$$

No later stable finish is empty. The entire update rule on the four stable states is pictured below, so the repeating claim is exhaustive, rather than inferred only from eleven trials.



Guide diagram, not to scale

K–1: Every arrow means add one at A and finish all sharing. Only the three states in the cycle occur after an addition. The initial empty state is outside the cycle.

Grades 2–3: The complete twelve-row table is below. Each rule reaches exactly $(1,0), (0,1), (1,1)$ after at least one addition. $(0,0)$ occurs only before the first addition.

Chips added	Add at A then finish	Add at B then finish
1	$(1,0)$	$(0,1)$
2	$(0,1)$	$(1,0)$
3	$(1,1)$	$(1,1)$
4	$(1,0)$	$(0,1)$
5	$(0,1)$	$(1,0)$
6	$(1,1)$	$(1,1)$
7	$(1,0)$	$(0,1)$
8	$(0,1)$	$(1,0)$
9	$(1,1)$	$(1,1)$
10	$(1,0)$	$(0,1)$
11	$(0,1)$	$(1,0)$
12	$(1,1)$	$(1,1)$

Why the cycles persist. The A-addition map is drawn on page 7. Interchanging the names A and B gives the B-addition map. Once the same pair recurs, the fixed rule repeats the same successors forever. The two rules traverse the three-cycle in opposite directions.

K–1: Ask whether an arrangement has appeared before. Once it has, ask what the same next action must do from that same arrangement.

Grades 2–3: An adult may turn a child's repeated rows into three state cards with arrows. Introduce the diagram after the repetition has a purpose.

Why sharing must stop on the four-cycle

Grades 4–5 P5 · return route

Idea. The score W = 3A + 4B + 3C drops by 2 at every legal share on the four-cycle and never goes below zero, so every run stops.

All three 12-chip starts stop. They happen to share one finish, although their share counts differ:

Start	Finish	Shares at each circle	Chips in sink
$(0,12,0)$	$(1,0,1)$	$(5,11,5)$	10
$(6,0,6)$	$(1,0,1)$	$(5,5,5)$	10
$(4,4,4)$	$(1,0,1)$	$(5,7,5)$	10

A finite score proves termination on this board. Give each chip at A or C a score of 3, each at B a score of 4, and each at the sink a score of 0. The total $W = 3A + 4B + 3C$ is a nonnegative integer. An A or C move removes two 3-point chips and creates one 4-point chip, decreasing W by 2. A B move removes two 4-point chips and creates two 3-point chips, also decreasing W by 2. Therefore infinitely many legal moves are impossible. This works for every finite start and every choice of legal moves on the four cycle. The three starts have scores 48, 36, 40; each finishes at score 6, so the total numbers of moves are 21, 15, 17.

First ask why "a chip falls into the sink every move" is false: B does not touch the sink. For a child who wants a numerical certificate, give the 3,4,3 weights and let them test each type of move.

Why every order gives the same piles and counts

Grades 4–5 P6 · return route

Idea. No legal run can be the first to exceed a finished run's share quota at a circle, so every complete run has the same share counts and hence the same final piles.

From $(0,8,0)$, the unique finish is $(1,0,1)$, with counts $(3,7,3)$ and 6 chips in the sink. Two full legal words are BBABBABCCBACB and BBCBBCBAABCAB. Each has 13 moves. They exchange A and C choices.

Why the share counts must agree. Fix one finished legal history, with u_v shares at each circle v. Imagine another legal history, and suppose it is about to be the first to exceed one of those quotas, at v. So far, v has shared exactly u_v times, and every neighbor has shared no more than its quota. Thus v has received no more chips than in the finished history and has sent exactly as many. Its present pile is at most its pile in that stable finish. It cannot legally share. This contradiction says no quota can be exceeded. If the other history also finishes, reverse the comparison; all quotas are equal. Equal counts determine equal final piles.

Child-facing proof conversation. Write the quotas 3,7,3 beside the sample board. Ask, “Could a different run be the first to make A share a fourth time? Before that happens, could A have received more chips than in our finished run?” Then discuss the same question for B and C. A careful explanation of that comparison is enough; symbolic notation is optional.